

Speech Processing

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Review of Signal Processing: LTI Systems

- A **Linear Time-Invariant (LTI)** system is completely characterized by its **impulse response** = response of the system to $\delta[n]$.
- Response of LTI system for any arbitrary input is

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

- Discrete time Fourier Transform of a signal is given as $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$

$$\omega = 2\pi \frac{F}{F_s} \quad -\pi \leq \omega \leq \pi$$

- ω in rad/sample; $\omega=0$ to π is considered as general frequency range (ignoring negative values $-\pi$ to 0).
- F_s : sampling rate in samples/sec or Hz
- F : continuous time frequency in cycles/sec or Hz

Frequency response

The **frequency response of LTI system** is the DTFT of the impulse response:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n} \quad \begin{array}{l} |H(e^{j\omega})| \quad (\text{Magnitude response}) \\ \angle H(e^{j\omega}) \quad (\text{Phase response}) \end{array}$$

Input-output relation: $y[n] = x[n] * h[n]$

$$Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})}$$

DTFT properties

Property	Time-Domain Operation	DTFT Relation
1. Delay (Time Shift)	$y[n] = x[n - n_0]$	$Y(e^{j\omega}) = e^{-j\omega n_0} X(e^{j\omega})$
2. Convolution	$y[n] = x[n] * h[n]$	$Y(e^{j\omega}) = X(e^{j\omega}) H(e^{j\omega})$
3. Symmetry (Time Reversal / Conjugation)	$y[n] = x[-n]$	$Y(e^{j\omega}) = X(e^{-j\omega})$
	$y[n] = x^*[n]$	$Y(e^{j\omega}) = X^*(e^{-j\omega})$

System function

System function $H(z)$ standard form

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

- **Zeros**: roots of the numerator
- **Poles**: roots of the denominator
- Poles and zeros are plotted in complex z -plane. $z = r^* e^{j\omega}$
- $|z|=r=1$ is called **unit circle** in z -plane

Frequency response from $H(z)$

$$H(e^{j\omega}) = H(z) \Big|_{z=e^{j\omega}}$$

- **Zeros near unit circle** → spectral **nulls (valleys)**
- **Poles near unit circle** → spectral **peaks (resonances)**

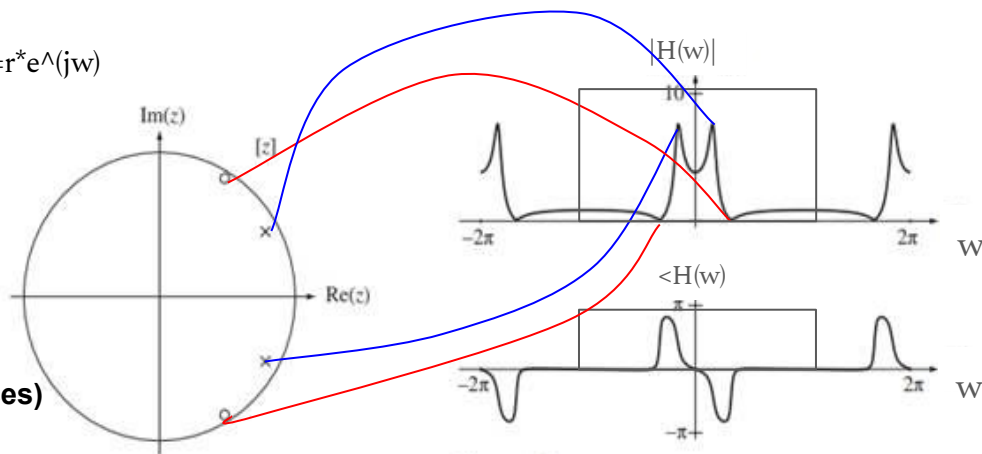


Figure 9.16

Pole-zero diagram of the transfer function,

$$H(z) = \frac{z^2 - 0.96z + 0.9028}{z^2 - 1.56z + 0.8109}$$

Figure 9.17

Magnitude and phase frequency response of the system whose transfer

$$\text{function is } H(z) = \frac{z^2 - 0.96z + 0.9028}{z^2 - 1.56z + 0.8109}$$

Autocorrelation of a signal

Auto correlation is similarity of a signal with its delayed-self.

Its a function of delay: $R_{xx}[k] = \sum_{n=-\infty}^{\infty} x[n] x^*[n - k]$

$R_{xx}[0] = \sum_{n=-\infty}^{\infty} |x[n]|^2$ gives energy of the signal.

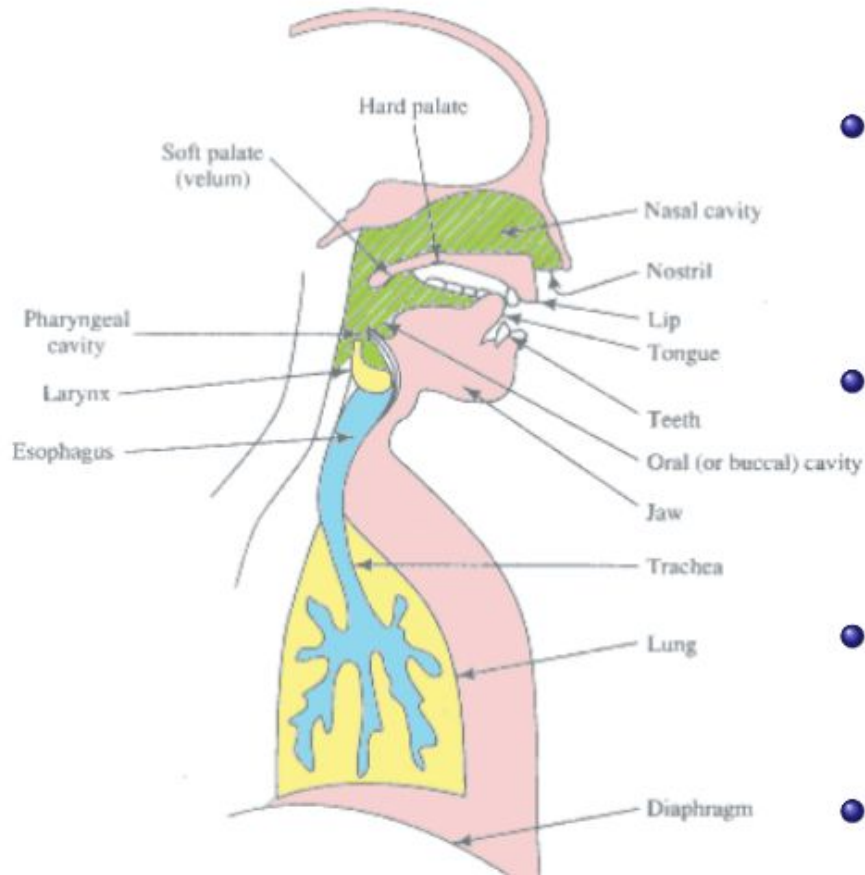
Autocorrelation is even symmetric function: $R[-k]=R[k]$.

Autocorrelation of a periodic signal is also periodic with same period.

DTFT of autocorrelation gives power spectral density.

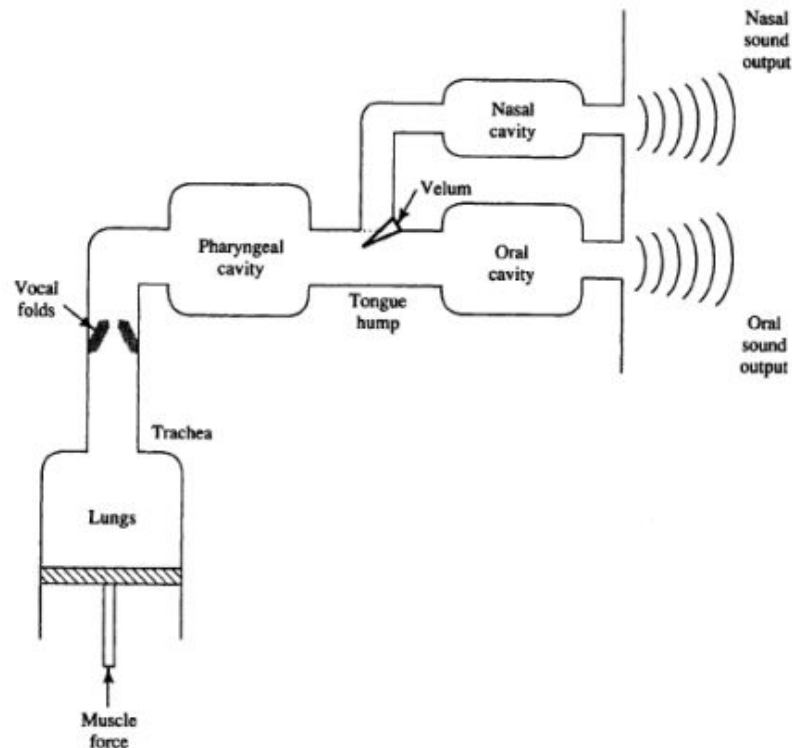
$$S_{xx}(e^{j\omega}) = \sum_{k=-\infty}^{\infty} R_{xx}[k] e^{-j\omega k}$$

Speech Production Mechanism



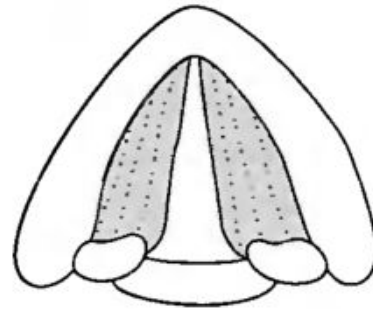
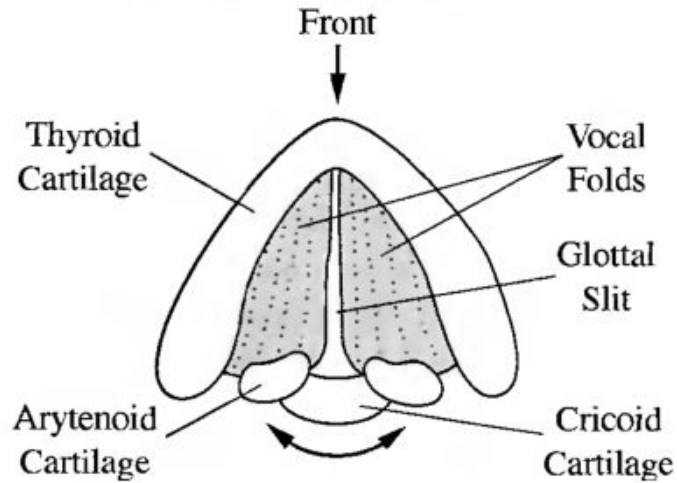
- Speech is the most sophisticated motor activity in the human body
- The motions are lightening fast and totally fluid, yet highly accurate
- Nasal cavity is much larger than oral cavity
- Animation

Block Diagram of Speech Production

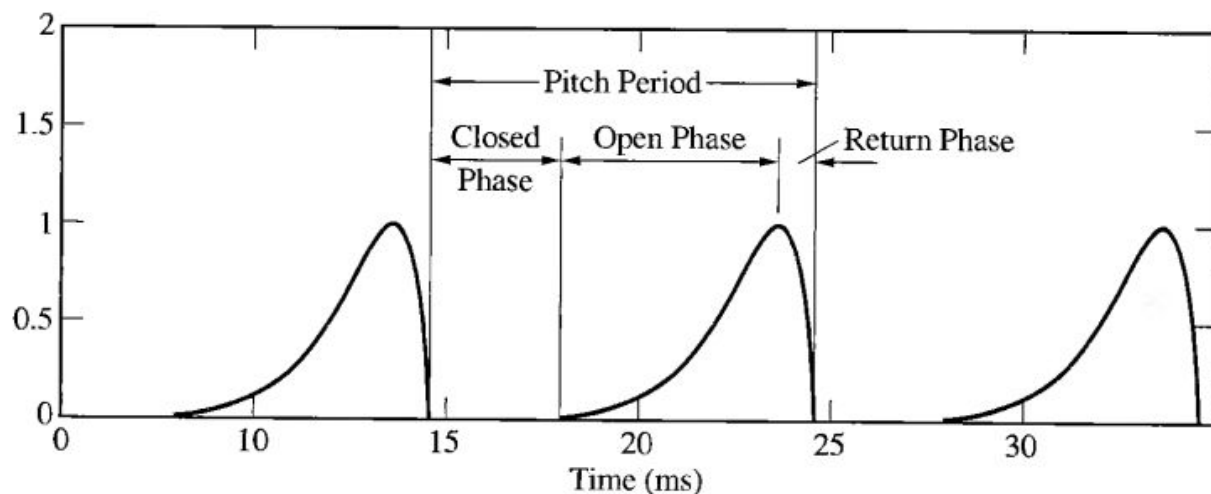


- Lungs act as source of energy
- Vocal folds chop the airflow from lungs into quasi periodic puffs of excitation
- Shape of the vocal tract determines the sound that is produced
- Velum is big enough is decouple nasal cavity, but not oral cavity

Vocal Folds - View & Operation



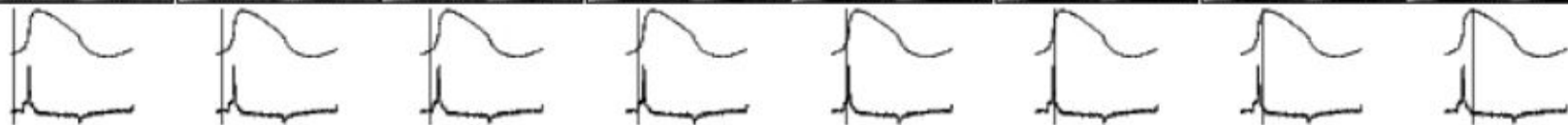
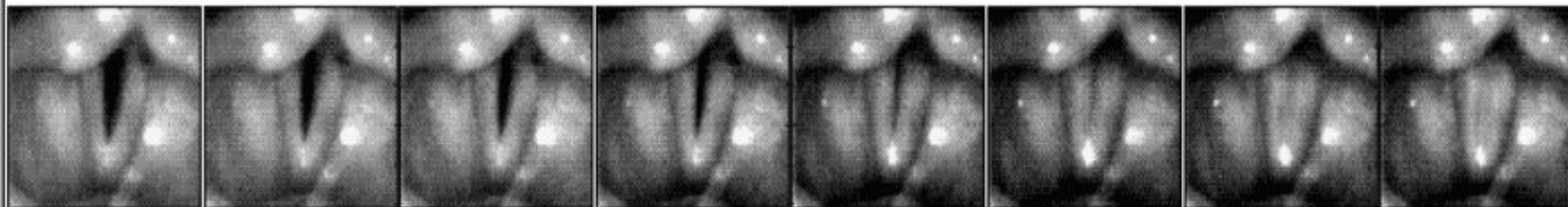
Glottal Flow



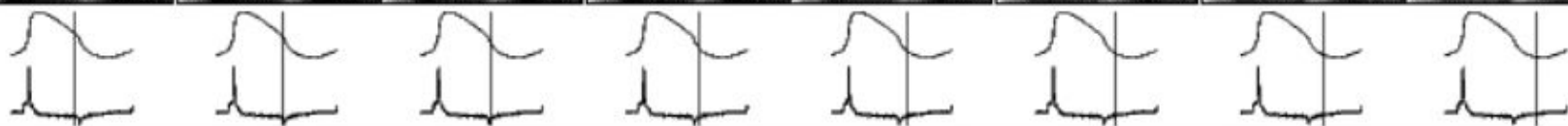
- Vocal folds are muscular tissues in the larynx
- Open when we breathe quietly and closed when we swallow
- Air coming from the lungs causes vocal folds to vibrate between open and closed positions when we talk
- Bernoulli Principle: Increase in the velocity of fluid occurs with simultaneous decrease in the pressure

Laryngeal Cycle

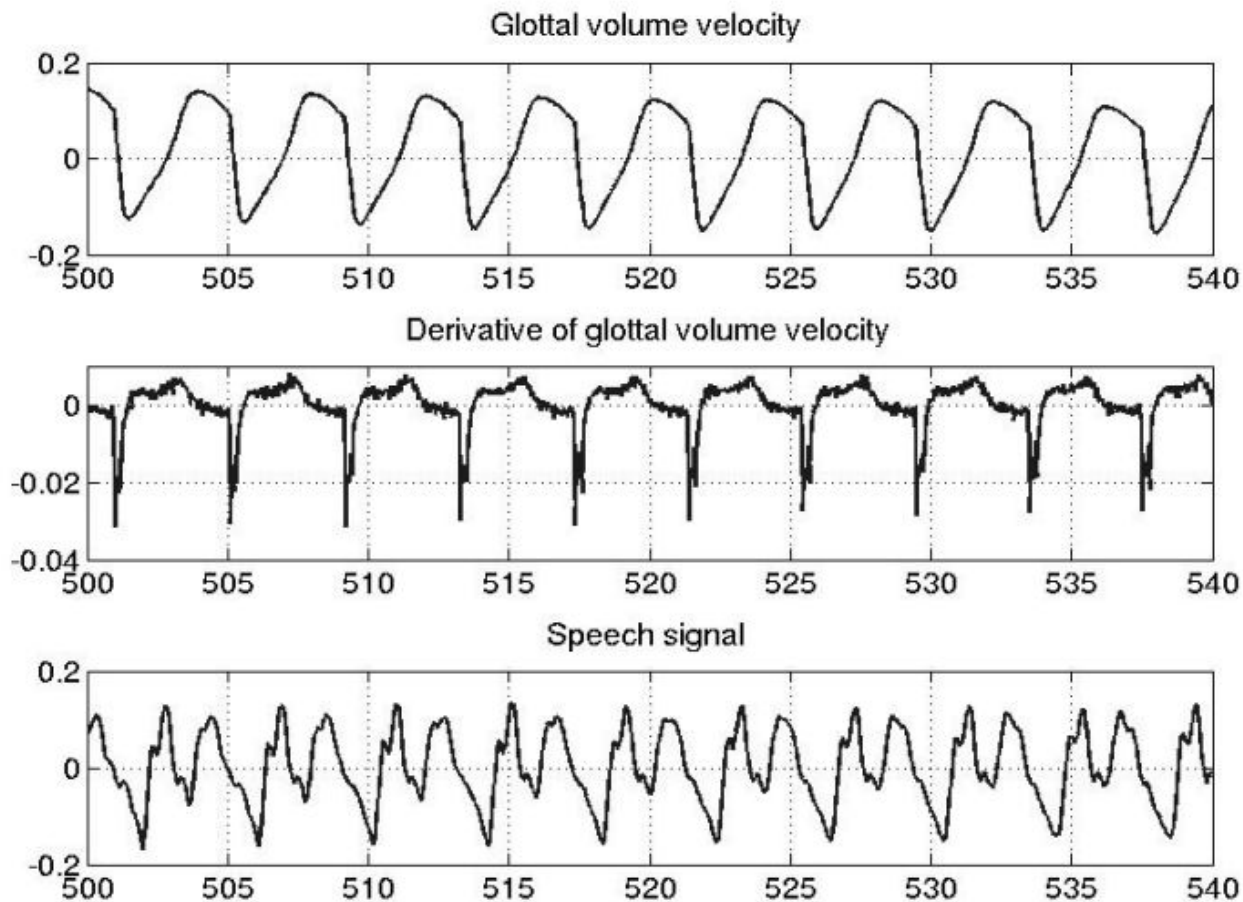
CLOSURE PHASE



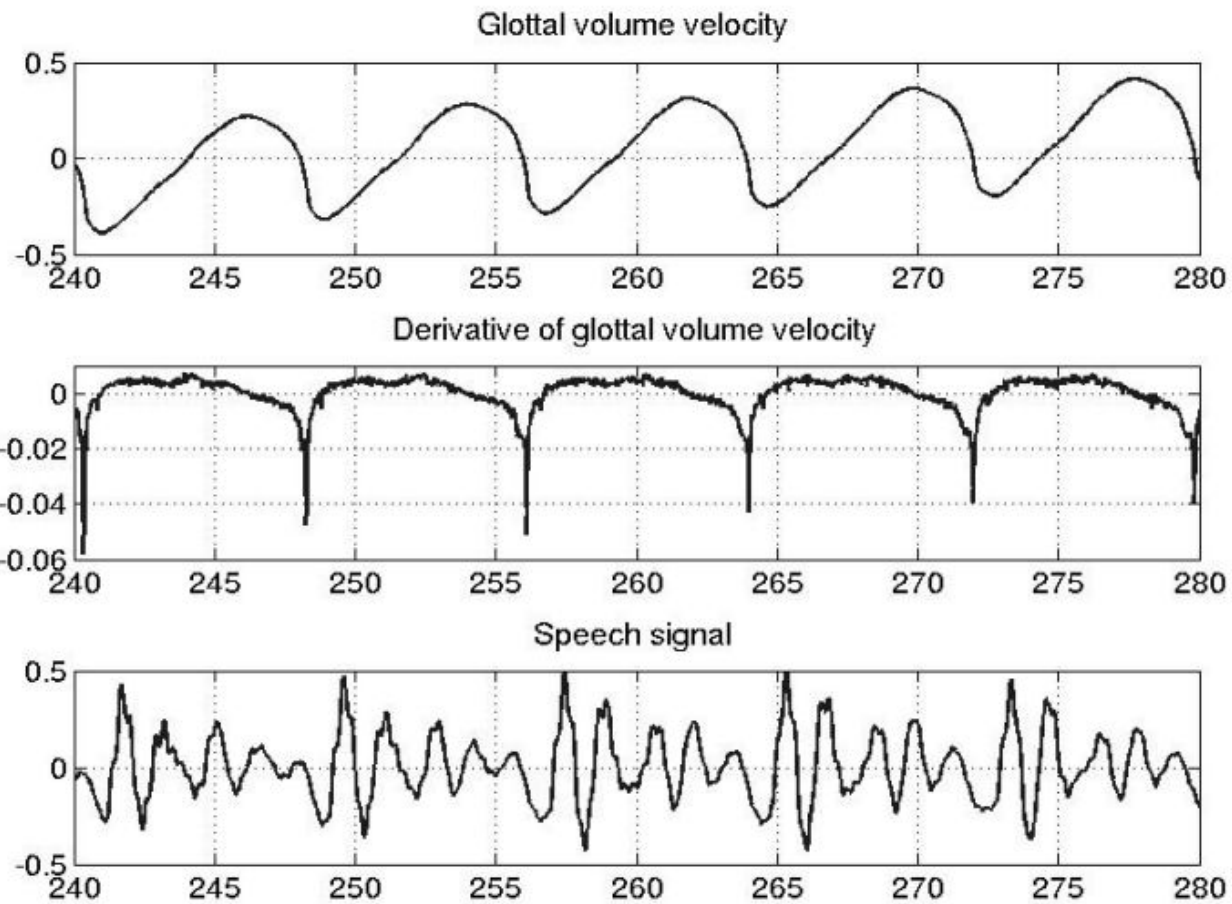
OPENING PHASE



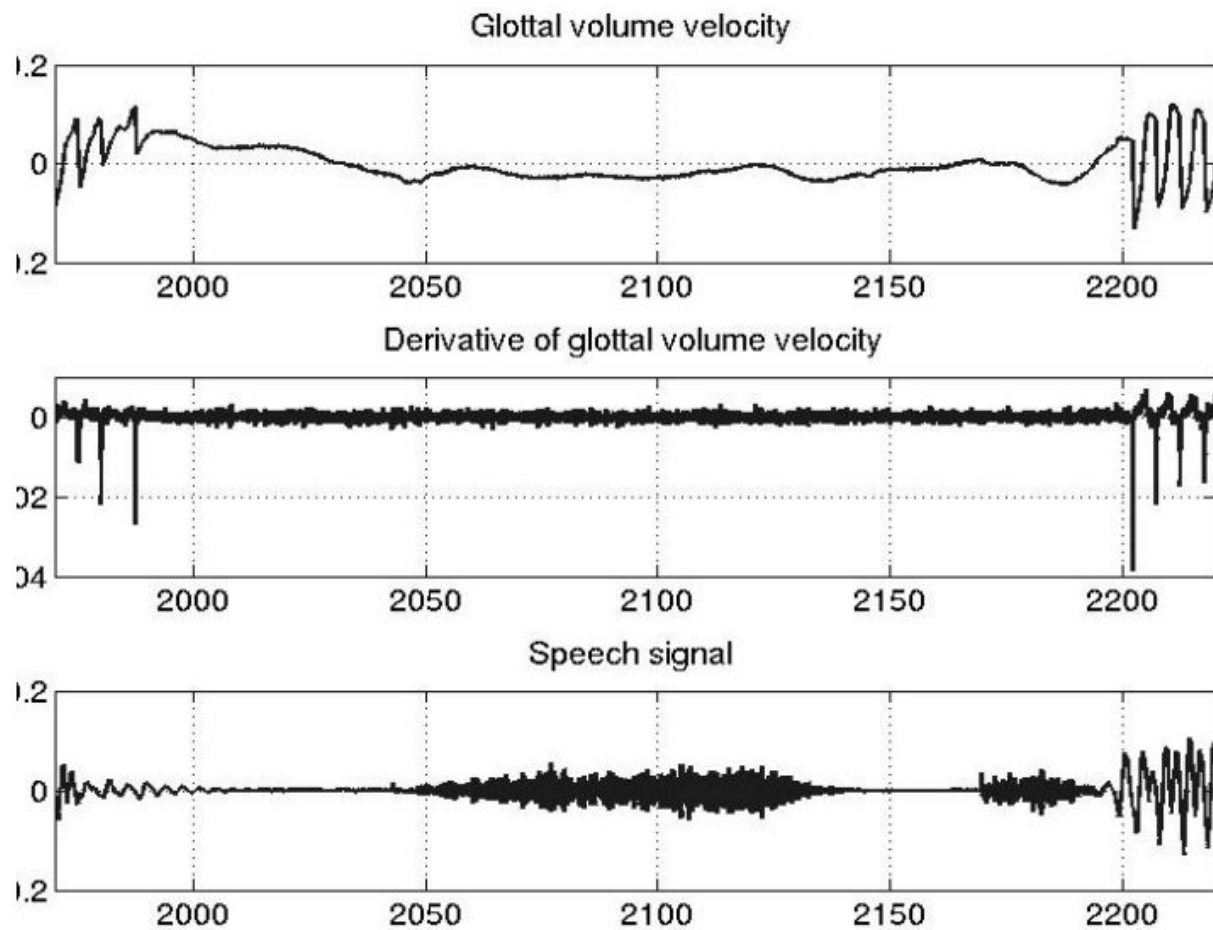
Voice Source - Female Speaker



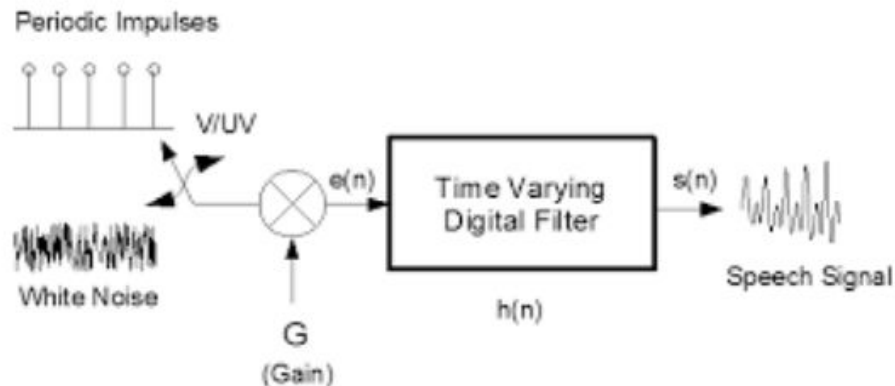
Voice Source - Male Speaker



Voice Source During Frication



Source-Filter Model of Speech Production



- Speech is an outcome of time-varying vocal tract filter driven by a time-varying excitation
- The state of the vocal cords, the positions, shapes and sizes of the various articulators – all change slowly over time, thereby producing the desired speech sounds
- Need to determine the state of VT from waveform - inverse problem